From prediction to planning

Making predictive analytics usable for health-system decision support in resource-limited settings

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Core argument: the bottleneck is not simply building better models - it is connecting prediction to governance, workflow, stakeholder interpretation and planning action.

OVERALL AIM

To examine how predictive analytics using longitudinal population and routine health data can be developed, interpreted and translated into actionable decision-support insights for health-system planning in resource-limited settings.

PROJECT OBJECTIVES

1. Map predictive analytics evidence using routine and population health data.
2. Identify gaps in implementation, governance, data integration and decision pathways.
3. Use stakeholder engagement to define decision-relevant case studies.
4. Apply predictive modelling and forecasting to longitudinal HDSS data.
5. Translate outputs into planning insights with health-system stakeholders.

WHY THIS MATTERS

Health systems generate large volumes of routine and population data, but planning often remains retrospective. Predictive analytics can help anticipate demand and risks - only if outputs are usable within real decision processes.

Key shift: from model performance to decision support.

STUDY SETTING

Navrongo HDSS, Ghana

A longitudinal population surveillance platform with repeated observations of births, deaths, migration, pregnancies and household context - suitable for person-time datasets, survival analysis, longitudinal modelling and forecasting.

EMPIRICAL DESIGN TO MEET THE GAP

Sequential, stakeholder-informed mixed-methods case study

1 Evidence Scoping review → **2 Priorities** Stakeholder case studies → **3 Models** HDSS prediction → **4 Forecast** Service pressures

5 Interpret Usability + barriers → **6 Translate** Decision-support framework

Outputs are aggregated for planning - not individual clinical decision-making.

EARLY DELIVERABLE: GLOBAL SCOPING REVIEW

Completed evidence mapping of predictive analytics for health-system decision support using routine or population data (2014-2025; n = 161).

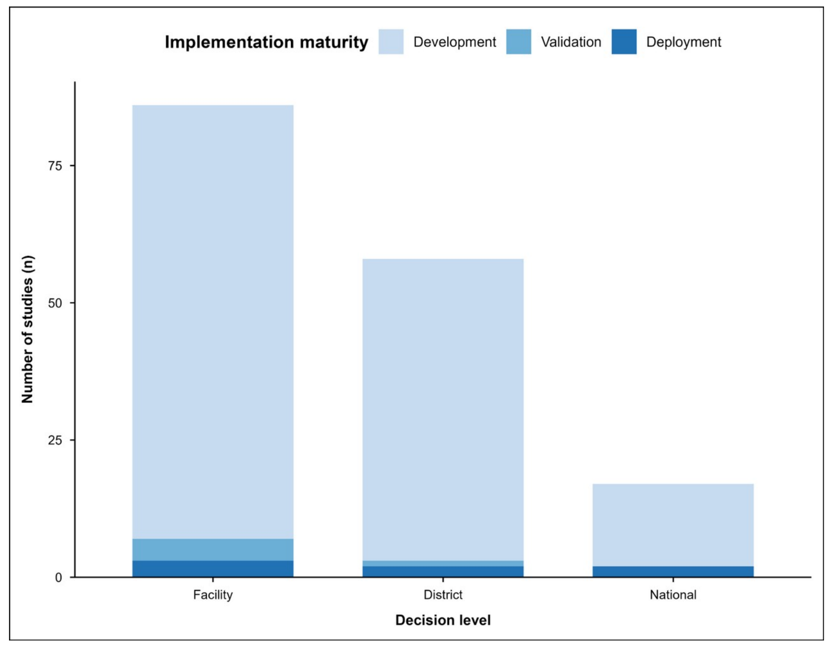
92.5%
development only
149/161 did not progress to validation or operational use

94.4%
no decision pathway
152/161 did not describe how outputs inform action

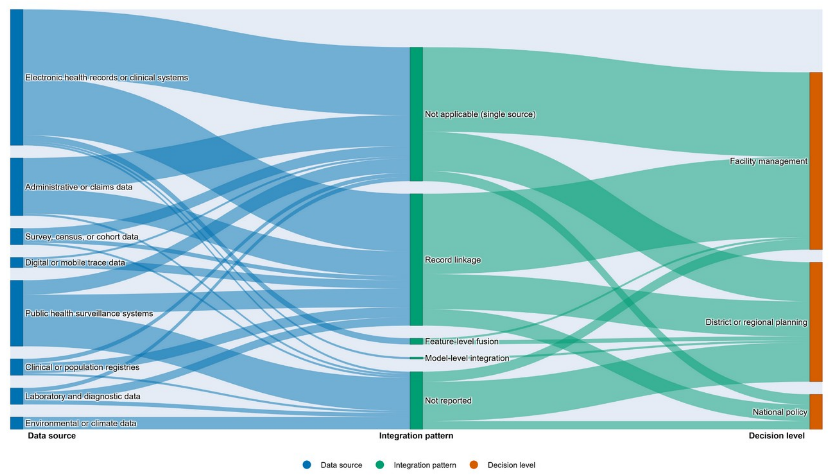
70.8%
uncertainty absent
114/161 did not report uncertainty information

1.9%
African studies
Only 3/161 studies were from the WHO African Region

What the review shows: predictive analytics is expanding, but the translation gap is institutional - governance, workflow integration, accountability and interpretation are usually missing.



Most predictive decision-support work remains at development stage across facility, district and national decision levels.



Data sources, integration patterns and decision levels are fragmented; linkage and decision pathways are often under-reported.

So what must change?

Data: Longitudinal + routine sources → Models: Prediction + forecasting → Meaning: Stakeholder interpretation → Action: Planning use

This PhD treats predictive analytics as an end-to-end decision-support process, not a stand-alone modelling exercise.

The poster therefore moves from overall project aims to the completed scoping review deliverable, the research gap it reveals, and the empirical design that addresses that gap.

RESEARCH GAP REVEALED BY THE REVIEW

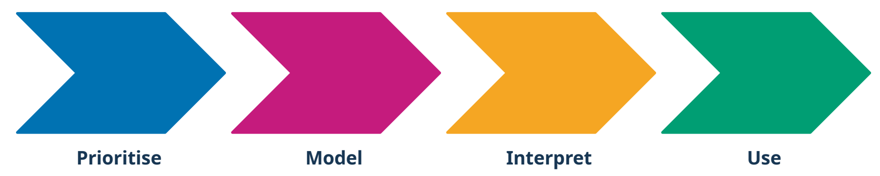
Existing work often demonstrates that models can predict, but rarely shows how predictions become governed, trusted, interpreted and used in routine planning.

Missing links:

- explicit decision pathways
- uncertainty communication
- governance and accountability
- workflow integration
- use of longitudinal population surveillance data

MY PHD RESPONSE

A stakeholder-informed pathway that starts with real planning priorities, uses longitudinal population data, and tests whether outputs are understandable, useful and actionable.



Potential case studies: under-five mortality and selected service-utilisation indicators, subject to stakeholder priorities and HDSS data suitability.

WHAT MAKES IT DIFFERENT

- Starts with decision needs, not just available variables.
- Uses Navrongo HDSS longitudinal structure to model change over time.
- Compares statistical, machine-learning and forecasting approaches for both performance and interpretability.
- Uses stakeholder interpretation to test whether outputs can support planning conversations.

PLANNED OUTPUTS AND IMPACT

- Peer-reviewed scoping review and empirical case-study paper.
- Stakeholder-defined health-system case studies.
- Predictive and forecasting outputs aggregated for planning.
- Decision-support framework for responsible predictive analytics.
- Ghana-facing policy/practice briefing, interactive dashboard and website for dissemination.

What I want people to remember

Prediction is only useful when it changes how health systems plan, allocate resources and learn.

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Selected sources: Dormechele, submitted scoping review; Oduro et al. 2012, Navrongo HDSS profile.